

Bingchen Wang

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Research Profile

My research studies intelligence under constraints: how human, artificial, and hybrid systems coordinate, represent, and decide under incentive, informational, and institutional limits. I organize the work along the lifecycle of an intelligent system—**development**: incentive design for collaborative machine learning (*Paid with Models*); **alignment**: preference reconstruction without demographic data or finetuning (*Prompts to Proxies*), with synthetic-panel testing ahead of costly deployment (SIM - PANEL); and **deployment**: strategic adaptation in AI-mediated hiring (ongoing). Across projects, I build methods that reduce the fixed cost of asking serious empirical questions about AI systems, relaxing dependence on abundant compute, sensitive data, proprietary access, or bespoke human pipelines.

Education

University of Oxford, St Hilda’s College

Oxford, UK

M.Phil. in Economics (*Distinction*)

Oct 2019 – Jun 2021

Thesis: *Detecting Parameter Shifts using Multiplicative-Indicator Saturation*.

Supervisors: David Hendry and Jennifer Castle

Columbia University

New York, USA

B.A. in Mathematics–Statistics (*Summa cum laude*)

Sep 2016 – Dec 2018

Phi Beta Kappa; Department Honors in Statistics; GPA: 4.13/4.33

City University of Hong Kong

Hong Kong SAR

B.Sc. (Hons.) in Computing Mathematics (*First Class Honours*)

Sep 2014 – May 2016

Joint Bachelor’s Degree Program with Columbia University; GPA: 4.13/4.3

Publications and Working Papers

Bingchen Wang, Zi-Yu Khoo, Jintan Wang. “Prompts to Proxies: Emulating Human Preferences via a Compact LLM Ensemble.” *arXiv preprint* arXiv:2509.11311, 2026. Under review

Bingchen Wang, Zhaoxuan Wu, Fusheng Liu, Bryan Kian Hsiang Low. “Paid with Models: Optimal Contract Design for Collaborative Machine Learning.” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 13, 2025, pp. 14176–14183. Oral presentation. doi:10.1609/aaai.v39i13.33552

Research Systems and Software

P2P — Preference reconstruction with compact LLM proxy ensembles

Pre-release (site)

Python; LLM APIs; survey-response pipelines; constrained Lasso/ElasticNet; held-out validation

- Built an inference-time framework for reconstructing population-level preferences from LLM proxy agents, without finetuning, sensitive demographic conditioning, or individual-level digital twins.
- Implemented the pipeline end to end: structured proxy generation, survey-response collection and cleaning, entropy-guided expansion, sparse aggregation, held-out validation, and automated reporting.
- Validated reconstruction fidelity on 14 waves of the American Trends Panel using held-out MSE and other distributional metrics, at an estimated cost under USD 1 per survey (Gemini-2.0-Flash).

SIM-PANEL — Synthetic panel simulation and benchmarking toolkit

Open source (GitHub)

Python; CLI/YAML; JSON/JSONL artifacts; schema validation; streaming ingestion; distributional diagnostics

- Built a CLI-driven, YAML-configured toolkit for generating, validating, and benchmarking synthetic panel-style datasets, runnable on a laptop with local models.
- Implemented randomized assignment, manual assignment, and agent self-selection policies for studying behavioral responses under different treatment-allocation and selection assumptions.

- Added source adapters, benchmark builders, comparison workflows, schema validation, automated reports, and distributional diagnostics against real or controlled reference data.

Research Experience

Independent Researcher

AI, Economics and Computational Social Science

Zibo, China

Jul 2025 – Present

- Originated and completed *Prompts to Proxies*, challenging digital-twin-style population simulation by reconstructing aggregate survey-response distributions from LLM proxy ensembles under weak information and compute constraints.
- Originated and am leading *Palimpsest*, a study of how AI-mediated screening and application-writing reshape labor-market signaling; wrote an accompanying research note on AI-mediated hiring, young people's agency, and labor-market equilibrium.
- Run the full research workflow independently: theory, implementation, cloud/API experimentation on small credit pools with retry handling around rate-limited APIs, empirical validation, documentation, paper writing, and public communication.

Research Assistant

Institute of Data Science, National University of Singapore

Singapore

Jul 2023 – Jun 2025

- Originated and completed *Paid with Models*, a contract-theoretic framework for incentivizing resource contribution in collaborative machine learning using models as rewards.
- Formulated the principal-agent problem under private contribution costs; transformed the non-convex optimal-contracting problem into a convex first-moment problem and designed a proportional assignment mechanism implementing its solutions in the original domain.
- Wrote the manuscript and rebuttal; designed the figures, poster, and slides; and delivered the paper as an oral presentation at AAAI-25.

Independent Learning and Research

Transition from Social Sciences to AI

Hong Kong & Zibo, China

May 2022 – Jun 2023

- Retrained from social science to machine learning: structured coursework in Python, machine learning, and deep learning; exploratory projects on text analysis, clustering, and policy simulation in PyTorch, published as open notebooks (*Learner's Corner*).

Senior Research Assistant

Research Hub on Institutions of China, The University of Hong Kong

Hong Kong SAR

Sep 2021 – Apr 2022

- Cleaned, linked, and analyzed large-scale Chinese survey datasets (CGSS, CLDS, CFPS); conducted exploratory data analysis and data-quality checks in R.
- Applied ordered logistic regression and grouped Lasso models to study associations among ideology, perceived income justice, and family background.
- Drafted empirical-analysis sections and supported manuscript evaluation through reviews of research design, data quality, econometric methods, and robustness checks.

Short-Term Consultant

The World Bank

Remote / Washington, DC, USA

Sep 2021

- Supported fiscal and economic analysis for the Jordan MTI team by updating debt/fiscal databases, oil-price monitors, Economic Monitor workflows, and Debt Sustainability Analysis materials.
- Automated recurring data updates in R, reducing processing time from over 30 minutes to under 5 minutes while improving reproducibility and reliability.

Research Assistant

Climate Econometrics Group, University of Oxford

Oxford, UK

Oct 2019 – Jun 2021

- Maintained and expanded large-scale datasets on anthropogenic CO₂ emissions, integrating data from NASA, FRED, OECD, and national sources; produced documentation and visualizations supporting reproducible dataset updates and empirical analysis.
- Conducted econometric analysis on parameter-shift detection and time-series diagnostics; work contributed to MPhil thesis on multiplicative-indicator saturation (MIS).

Presentations

Invited Talk: “Let’s Talk About Trust — Paid with Models,” Trusted CollabML Lab End-of-Project Meeting, Singapore, Mar 2025.

Workshop Presentation: “Fine-Tuning LLMs with Noisy Data for Political Argument Generation,” Good-Data Workshop, AAAI-25, Philadelphia, USA, Mar 2025. Presented on behalf of S. Churina and K. Jaidka.

Oral Presentation: “Paid with Models: Optimal Contract Design for Collaborative Machine Learning,” AAAI-25 Conference, Philadelphia, USA, Mar 2025.

Teaching

Teaching Assistant, Department of Mathematics, Columbia University, New York, NY, 2017–2018.

Undergraduate Tutor, Department of Statistics, Columbia University, New York, NY, 2017–2018.

Program Coordinator, Wall Street Training Program, GEC Academy, Beijing, China, Jul–Aug 2017.

Public Writing

South China Morning Post

“Hong Kong’s Covid-19 testing regime must be refined so that it’s not all stick, no carrots,” Feb 4, 2021.

“Why death of George Floyd should make the world take a good look at itself,” Jun 14, 2020.

“For Hong Kong, the only way out of a prisoner’s dilemma is to give and take,” Nov 29, 2019.

Awards and Honors

Joseph and Norma Preziosi Endowed Scholarship, Columbia University, 2018

Dean’s Scholarship, College of Science and Engineering, City University of Hong Kong, 2017

HKSAR Government Scholarship, City University of Hong Kong, 2015–16

Chan Feng Men-ling Chan Shuk-lin ELC Scholarship, City University of Hong Kong, 2014

Full Tuition Scholarship, City University of Hong Kong, 2014, 15, 17

Technical Skills

Programming: Python, R, MATLAB, Stata, OxMetrics, Git; pandas, NumPy, scikit-learn, PyTorch

Tools: LaTeX/Overleaf, Notion, VS Code, Google Cloud, Alibaba Cloud, LLMs (coding/debugging)

Software Development: CLI design, YAML configuration, JSON/JSONL artifacts, schema validation, streaming data ingestion, parallel processing, API-based experimentation, token/cost tracking, automated reporting, Markdown/Sphinx documentation

Languages: English, Mandarin Chinese, Cantonese (intermediate)